

SIMATS ENGINEERING
ASSESSMENT TOOL 2 : Scenario-Based

(34)

CO2- AT2
Slot D'

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1510

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

Code of Conduct

I Abdul Razak Shaik (Name / Reg No) 192524358 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Abdul Razak Shaik

① Cloud Architecture for video Streaming platform :-

Architecture Designs:-

- ① Use a Content Delivery Network (CDN) to cache videos closer to user.
- ② Store videos in cloud object storage
- ③ Deploy application server behind a Load Balancer.
- ④ Enable Auto Scaling Groups to add/remove servers automatically.
- ⑤ Use a distributed database and monitoring service.

Load Balancing and Auto Scaling

① Load Balancer

- Distributes user requests evenly across servers
- Prevents server overload
- Improves availability

② Auto Scaling:

- Automatically launches new instances during high traffic.
- Removes unused instances during low traffic.
- Ensures smooth streaming with minimal buffering.

Cost Optimizations

- Use CDN caching to reduce bandwidth costs.
- Use reserved instances for predictable workloads
- Use spot instances for background processing
- Enable auto scaling to avoid paying for idle resources
- Compress videos and optimize storage lifecycle policies

Conclusions:-

CDN + Load Balancer + Auto Scaling provides high performance scalability and cost efficiency.

②. Multi-cloud Container Deployment:-

Proposed Solution:-

- Use Kubernetes for Container Orchestration
- Deploy clusters across multiple cloud providers
- Use Docker containers for application packaging
- Store Container images in a shared container registry.

Service Discovery and Scaling:-

- Kubernetes service automatically discovers containers
- Internal DNS routes traffic to available pods.
- Horizontal pod autoscaler increased or decreased pods based on CPU or Memory usage.

Benefits:-

- High availability
- Vendor independence
- Disaster recovery
- Better geographic coverage

Risks:-

- Increased management complexity
- Higher networking costs
- Security policy differences
- Data synchronization challenges

Conclusion:-

Kubernetes with a multi-cloud strategy provides consistent deployment, scalability and resilience.

③

Backup and Disaster Recovery for e-learning Platform

Solution Designs:-

- Store data in cloud object storage with versioning Enabled.
- Enable automatic backups
- Replicate data to another region
- Use snapshots for virtual machines and databases.

Achieving RTO and RPO:-

• RTO (Recovery Time Objective):

- use automated failover
- Keep standby resources ready
- Enables quick service restoration.

• RPO (Recovery point objective):

- Continuous replication minimizes data loss
- Frequent backups reduce recovery gaps.

Business Continuity Steps:

1. Enable automatic daily backups.
2. Replicate data across regions.
3. Test disaster recovery regularly.
4. Automate failover procedures
5. Continuously monitor backup status.

Hybrid Cloud for Government Agency:

Hybrid Architecture:

- Sensitive Applications remain on-premises
- Non-Sensitive Applications run in the public cloud.
- Connect both environments using a secure VPN or dedicated connection.
- Use centralized management for monitoring.

Identity Management:

- Implement Single Sign On (SSO)
- Use Multi Factor Authentication (MFA)
- Apply Role-Based Access Control (RBAC)

Encryption:

- Encrypt data during transmission using TLS.
- Encrypt stored data using AES-256
- Secure Encryption Keys using a Key Management System.

Challenges:

- Complex management, Compliance requirements
- Network latency, Data Synchronization
- Increased Operational Cost.

Conclusion: A hybrid cloud balances security, compliance, and scalability while protecting sensitive government data.

⑤ High Availability and Fault Tolerance for SaaS

Multi-Region Deployment Strategy :-

- Deploy application servers in multiple regions
- Use a global balancer to route traffic
- Maintain active or standby deployments.
- Replicate databases across regions

Data Replication and Failover :-

- Enable Synchronous replication within a region
- Use asynchronous replication across regions.
- Configure automatic failover to the healthy region if one region fails.
- Regularly test failover mechanisms.

Monitoring and Alerting :-

- Monitor CPU, memory, storage and network usage.
- Track application response time and errors.
- Configure alerts for failures and unusual activity.
- Use centralised logging and dashboards.

Conclusion:- Multi-region deployment with replication, automatic failover and continuous monitoring ensures high availability and uninterrupted service.